5CM510 – Network System development
Lecture 13: Internet of Things

Lecture Outline

- Introduction to Internet of Things (IoT)
- IoT architecture and communication technologies
- IoT protocols, sensors and actuators
- IoT applications and use cases

Section 1

Introduction to Internet of Things (IoT)

What is Internet of Things (IoT)

- The Internet of Things (IoT) encapsulates a vision of a world in which billions of objects with embedded intelligence, communication means, and sensing and actuation capabilities will connect over IP (Internet Protocol) networks.
- IoT can be defined as a system where multiple objects communicate with each other and share data through sensors and digital connectivity.
- Machine-to-machine (M2M) solutions are comprised of linear communication channels between the machines to make them work in a cycle. Here, the action of one machine triggers the activity of other.
- The term Internet of Things was first used by Kevin Ashton in 1999 that was working in the field of networked RFID (radio frequency identification) and emerging sensing technologies.
    - IoT was “born” sometime between 2008 and 2009. In 2010, the number of everyday physical objects and devices connected to the Internet was around 12.5 billion.
    - Nowadays there are over 25 billion devices connected to the IoT - A smart device per person.
- The number of smart devices or “things” connected to the IoT is expected to increase to a further 50 billion by 2020.

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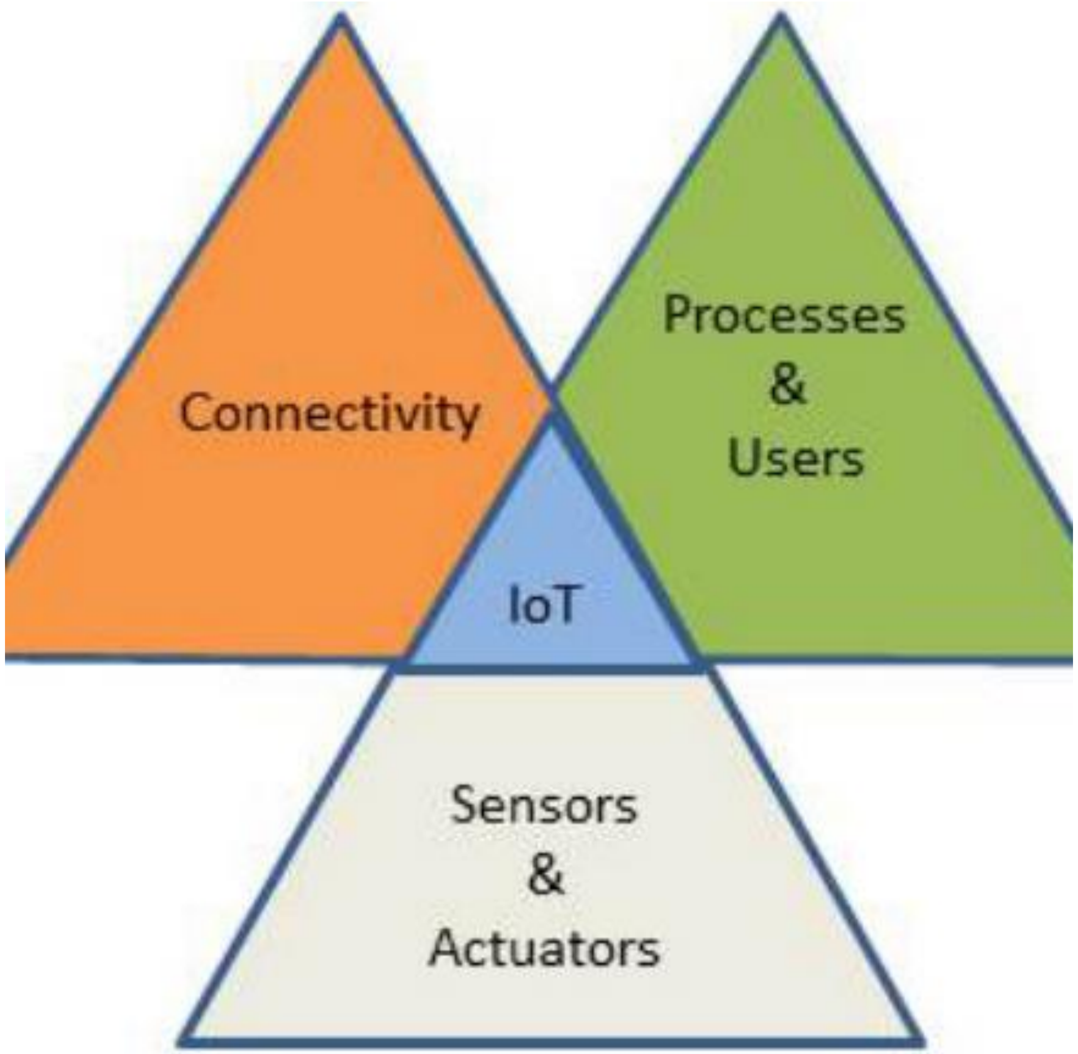
The IoT Concept

- The IoT can be seen as a combination of sensors and actuators providing and receiving information that is digitalized and placed into bidirectional networks able to transmit all data to be used by a lot of different services and final users.
- Multiple sensors can be attached to an object or device in order to measure a broad range of physical variables or phenomena and then transmit all data to the cloud.
- The sensing can be understood as a service model.



Internet of Things (IoT) Symbolic View

Section 2

IoT Architecture and Communication Technologies

Layered Architectural design

Four-layer architecture of IoT	
Object Sensing Layer	Sensing the physical objects and obtaining data.
Data Exchange Layer	Transparent transmission of data through communication networks.
Information Integration Layer	Processing of the uncertain information acquired from the networks, filtering undesired data and integratation of main information into usable knowledge for services and final users.
Application Service Layer	Provides content services to users.

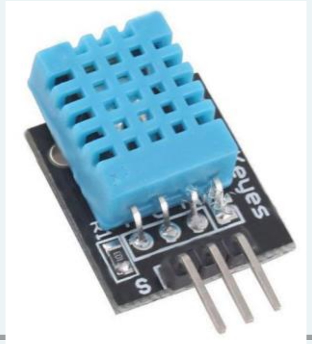
- The architecture of IoT systems can be divided into four layers: Object sensing layer, data exchange layer, information integration layer, and application service layer

Communication technologies (Link layer)

- Short-Range Wireless Technology: Bluetooth low-energy (BLE) compatible mesh network ,Light-Fidelity (Li-Fi), Near-field communication (NFC) - a range of 4 cm, QR codes and barcodes, Radio-frequency identification (RFID), Wi-Fi (IEEE 802.11 standard), ZigBee (IEEE 802.15.4 standard).
- Medium-Range Wireless Technology: HaLow (variant of the Wi-Fi standard), LTE-Advanced (Long term evolution at high data rate)
- Long-Range Wireless Technology: Low-power wide-area networking (LPWAN), Very small aperture terminal (VSAT)
- Wired Technology: Ethernet, Multimedia over Coax Alliance (MoCA).

IoT components

- Thing: IoT is interconnected with various sensors to collect the data and actuators to perform actions corresponding to the commands received from the cloud. e.g., DHT11, Nest Thermostat
- Gateway: It is used for data filtering, preprocessing, and communicating it to the cloud and vice versa (receiving the commands from the cloud). e.g., Raspberry Pi 5
- Cloud gateway: It is used to transmit data between the gateways and IoT central servers (ThingSpeak).
- Streaming data processor: It distributes the data coming from sensors to the relevant devices connected in network.



Sensitivity: Internal

Addressing the multitude of IoT devices: IPv4 to IPv6?

- When we use the Internet for any activity (e-mail, data transmission, web browsing, downloading files) or any other service or application, communication between different network elements and our own computer, laptop or smart phone, uses an Internet protocol (IP).
    - The IP which specifies the technical format of packets and the addressing scheme for computers to communicate over a network.
- The first version of IP publicly used was IPv4 (Internet protocol version 4). This protocol was created by the Defense Advanced Research Projects Agency (DARPA).
    - The use of addresses of 32 bits length limits the total number of possible addresses to approximately 4.3 billion addresses for devices connected to internet around the world.
    - The number of devices connected to Internet will soon overtake than the number of addresses provided by IPv4.
- The agency responsible for standardization of Internet protocols: The IETF (Internet Engineering Task Force) has been working in a new IP version from 1998: The IPv6, the successor protocol that is intended to replace IPv4 was first formally described in Internet standard document RFC 2460.

IPv6 and 6LoWPAN (Internet Layer)

- IPv6 will coexist with the older IPv4 for some time. The deployment of IPv6 will be made gradually in an orderly coexistence with IPv4.
- Client devices, network equipment, applications, content and services are to be adapted to the new Internet protocol version IPv6.
- The transition from IPv4 to IPv6 will establish a common set of standards between companies, educational systems, around the world.
- IPv6 uses a 128-bit address format, allowing 2^{128}, or approximately 3.4 $*10^{38}$ addresses, approximately 8 1028 times as many as IPv4.
- While increasing the pool of addresses is one of the most important benefits of IPv6, there are other important technological changes in IPv6 that will improve the IP protocol: easier administration, better multicast routing, a simpler header format and more efficient routing, built-in authentication and privacy support among others.
- 6LoWPAN: 6LoWPAN stands for IPv6 over Low-power Wireless Personal Area Networks. It is an extended layer for IPv6 through IEEE802.15.4 links. It operates on 2.4 GHz of frequency with a data transmission rate of 250 kbps.

Structure of IPv6 Header	
Version	4-bit Internet Protocol version number = 6.
Traffic Class	8-bit traffic class field.
Flow Label	20-bit flow label.
Payload Length	16-bit unsigned integer. Length of the IPv6 payload, i.e., the rest of the packet following this IPv6 header, in octets.
Next Header	8-bit selector. Identifies the type of header immediately following the IPv6 header. Uses the same values as the IPv4 protocol field
Hop Limit	8-bit unsigned integer. Decremented by 1 by each node that forwards the packet. The packet is discarded if Hop Limit is decremented to zero.
Source Address	128-bit address of the originator of the packet
Destination Address	Address 128-bit address of the intended recipient of the packet (possibly not the ultimate recipient, if a routing header is present).

IPv6 Header format

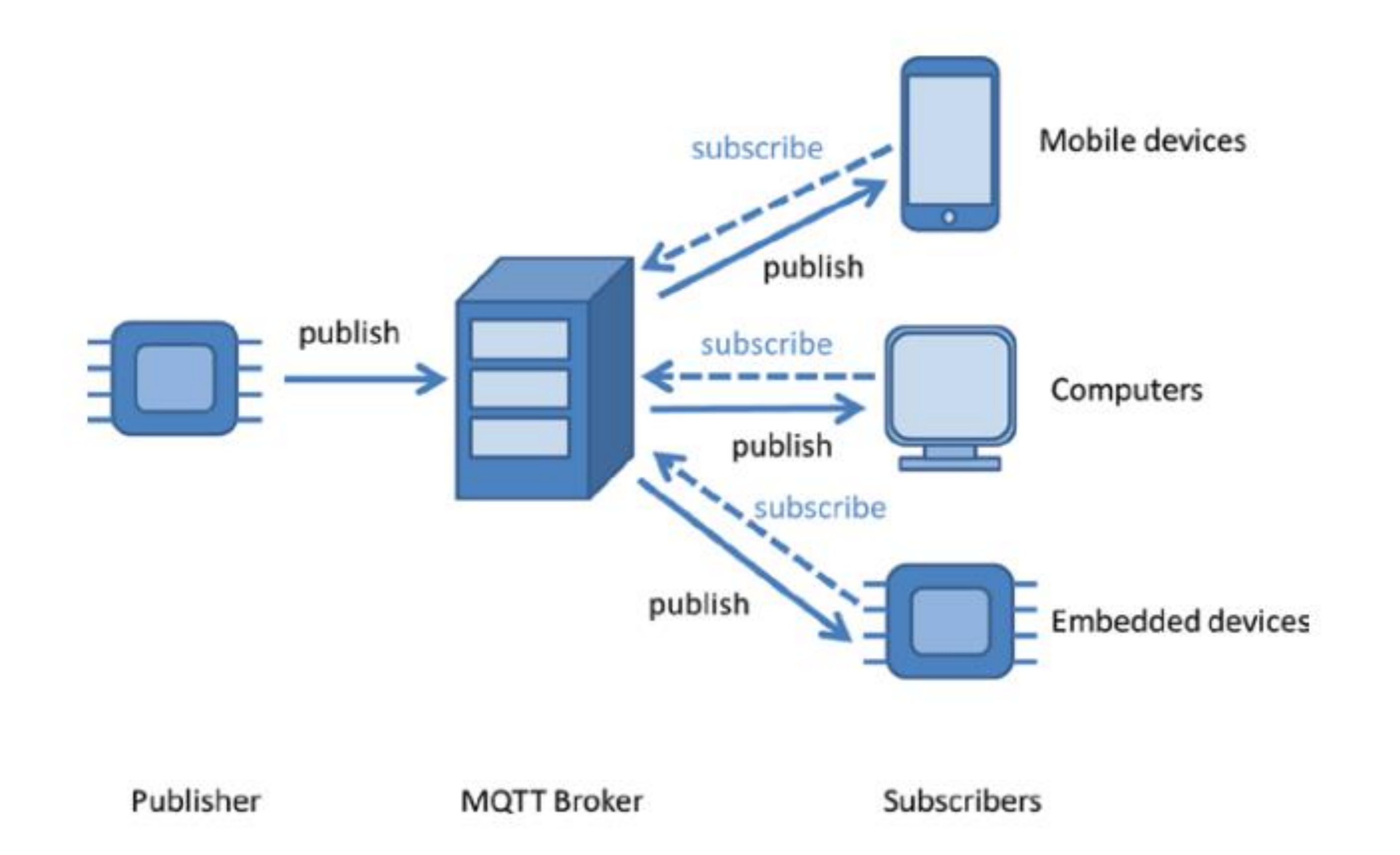
Section 3

IoT Protocols, Sensors and Actuators

OSI Model for IoT Protocols

Application layer	RES TAPI, JSON-IPSO objects, Binary objects
Transport layer	CoAP, MQTT, XMPP, AMQP, LLAP, DDS, SOAP, UDP, TCP, DTLS
Internet layer	6LoWPAN, IPv6, uIP, NanoIP
Link layer	IEEE802.15.4, IEEE802.11, ISO/IEC 8092:2004, NB-IoT, EC-GSM-IoT, Bluetooth, ANT, ISA100.11a, EnOcean, LTE-MTC
Physical layer	Devices, objects, things

Transport Protocol: MQTT (MQ Telemetry Transport)



Other Transport/Application Layer protocols

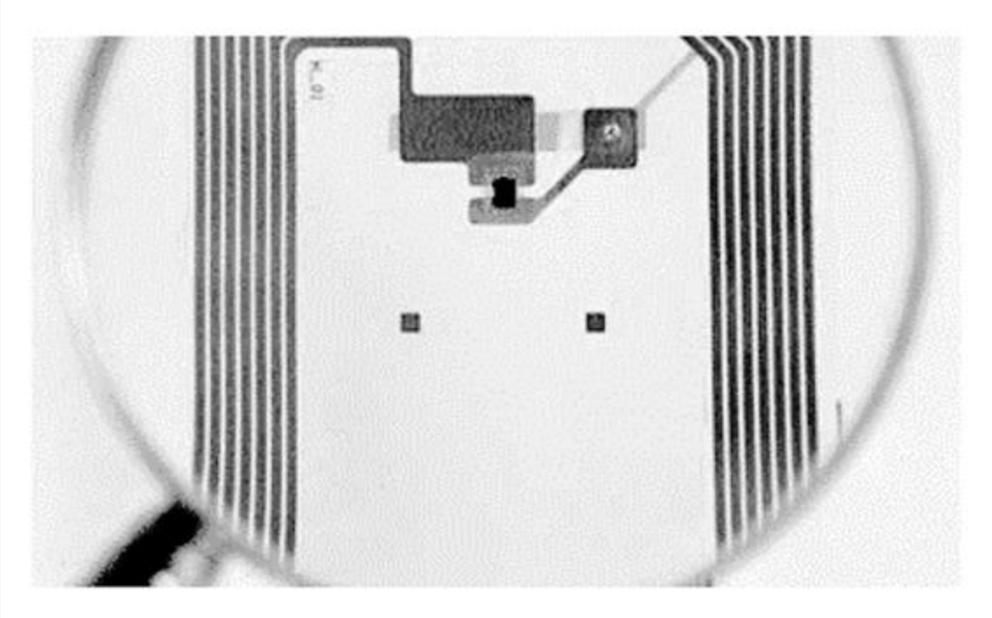
- DTLS (Datagram Transport Layer): The DTLS protocol provides the communication privacy for datagram protocols. It is used for prevention of tampering, message forgery, or eavesdropping in a network.
- NanoIP: Nano Internet Protocol is meant to establish communication among embedded sensors and devices without the overhead of TCP/IP.
- CoAP (Constrained Application Protocol): CoAP is an application layer protocol. It is designed to translate to HTTP for simplified integration with the web.
- REST: It stands for Representational State Transfer.
- SOAP: It stands for Simple Object Access Protocol.

Sensor Classification	
Sensor Data Providers	Business entities that deploy and manage sensors by themselves.
Organizations	Public or Private. Public infrastructures. Commercial organizations. Private corporations: Technology and services providers.
Personal and Households	Mobile phones, smart watches, gyroscopes, cameras, GPS, accelerometers microphones, laptops, food items and household items, such as televisions, cameras, freezers, microwave furnaces, washing machines, smart appliances etc

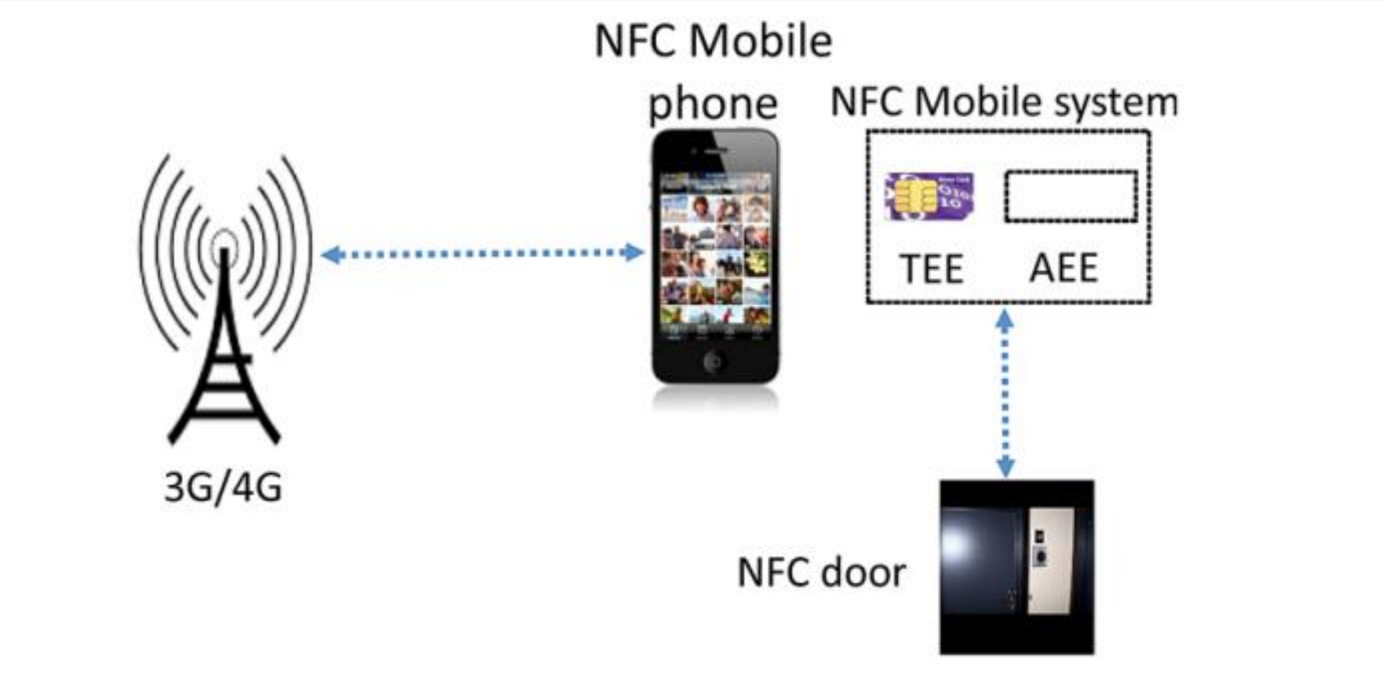
Smart connected products capabilities	
Monitoring	The external environment. The product's condition, operations and usage.
Control	Product functions control. Personalization of the user experience. Programing.
Optimization	Predictive diagnostics. Product performance optimization. Costs reduction.
Autonomy	Autonomous product enhancement and personalization. Self-diagnosis and repair. Coordination operation with other products
Efficient decision making process.	Real-time data for decision making.

Sensor classification and capabilities

Sensors: RFID Tags (Active and Passive)

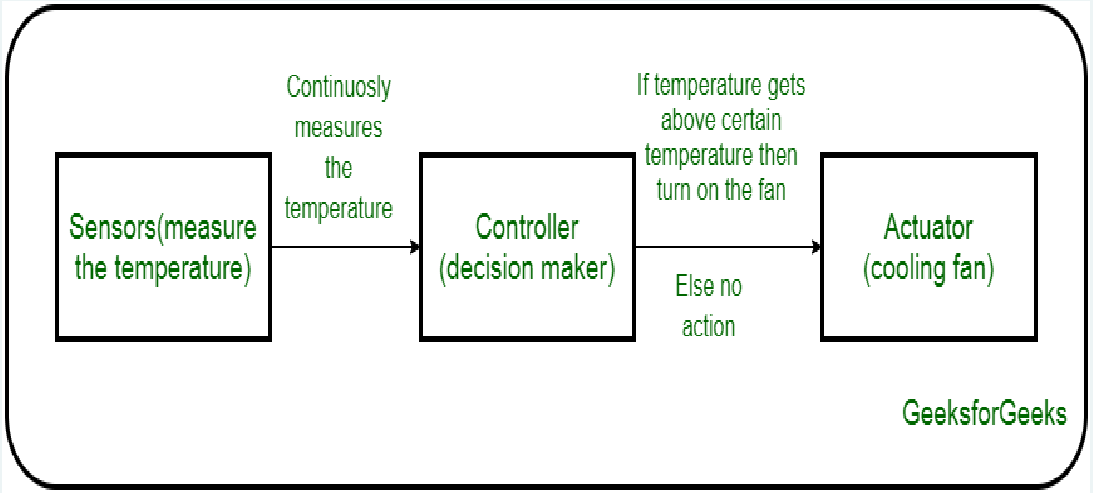


Sensors: NFC-enabled Mobile keys



Actuators

- An actuator is a machine component or system that moves or controls the mechanism of the system.
- Hydraulic Actuators – A hydraulic actuator uses hydraulic power to perform a mechanical operation. They are actuated by a cylinder or fluid motor.
- Pneumatic Actuators – A pneumatic actuator uses energy formed by vacuum or compressed air at high pressure to convert into either linear or rotary motion.
- Electrical Actuators – An electric actuator uses electrical energy, is usually actuated by a motor that converts electrical energy into mechanical torque.

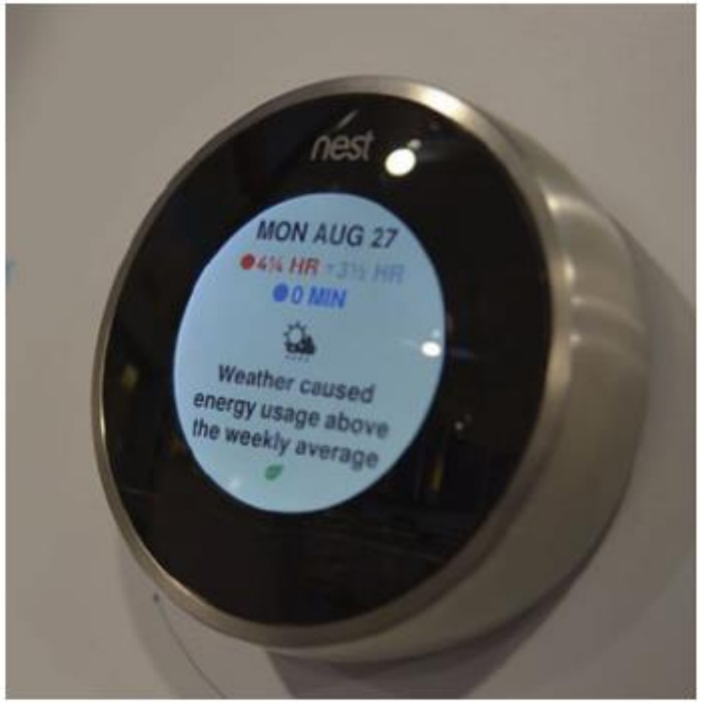


Actuators in IoT - GeeksforGeeks

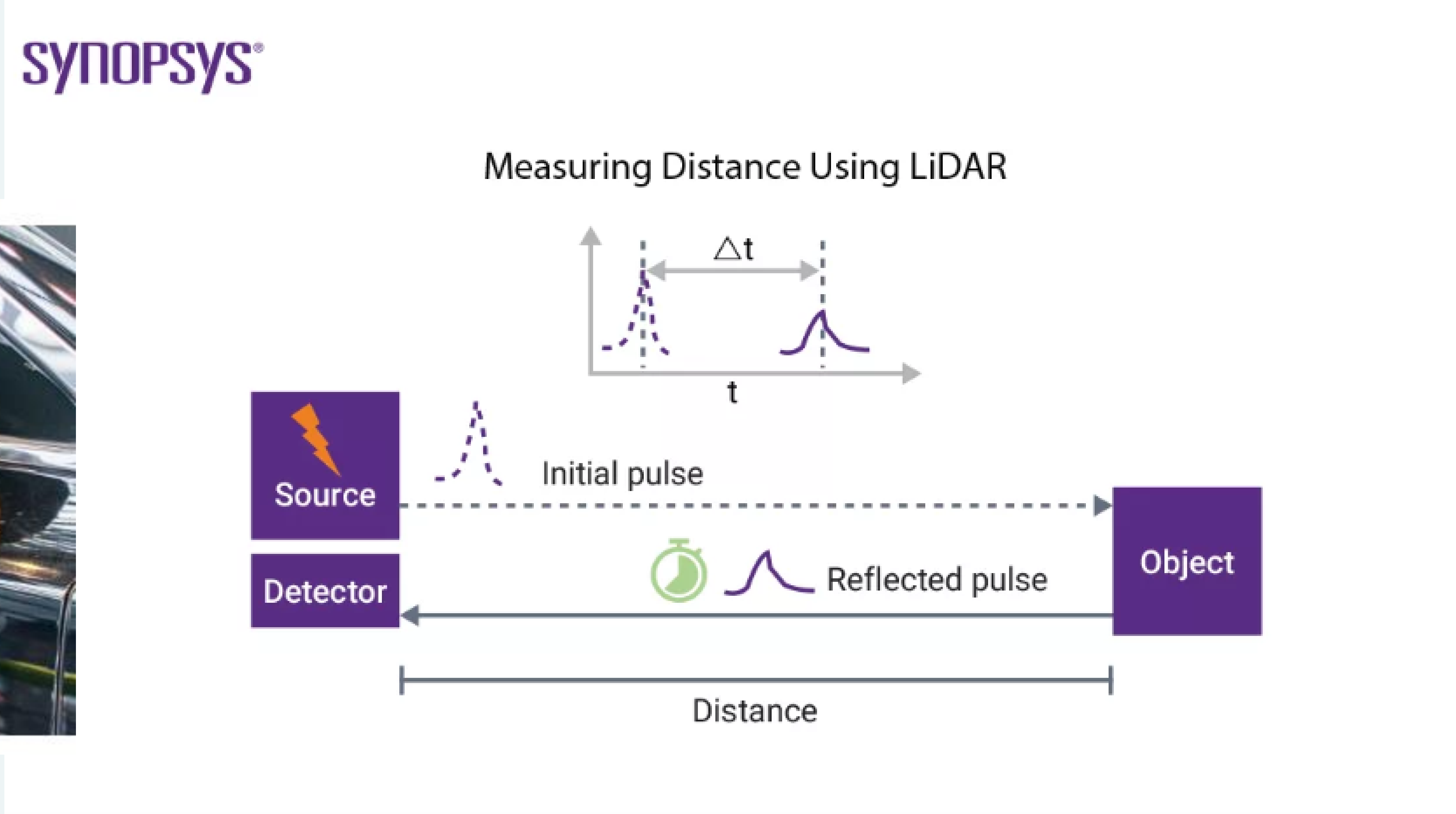
Section 4

IoT Application and Use cases (for Industrial IoT)

Energy: Landis Smart Meter and Nest Thermostat



LiDAR sensor– Self-driving cars and Safety



[What is LiDAR and How Does it Work? | Synopsys]

Healthcare: Body Area Network

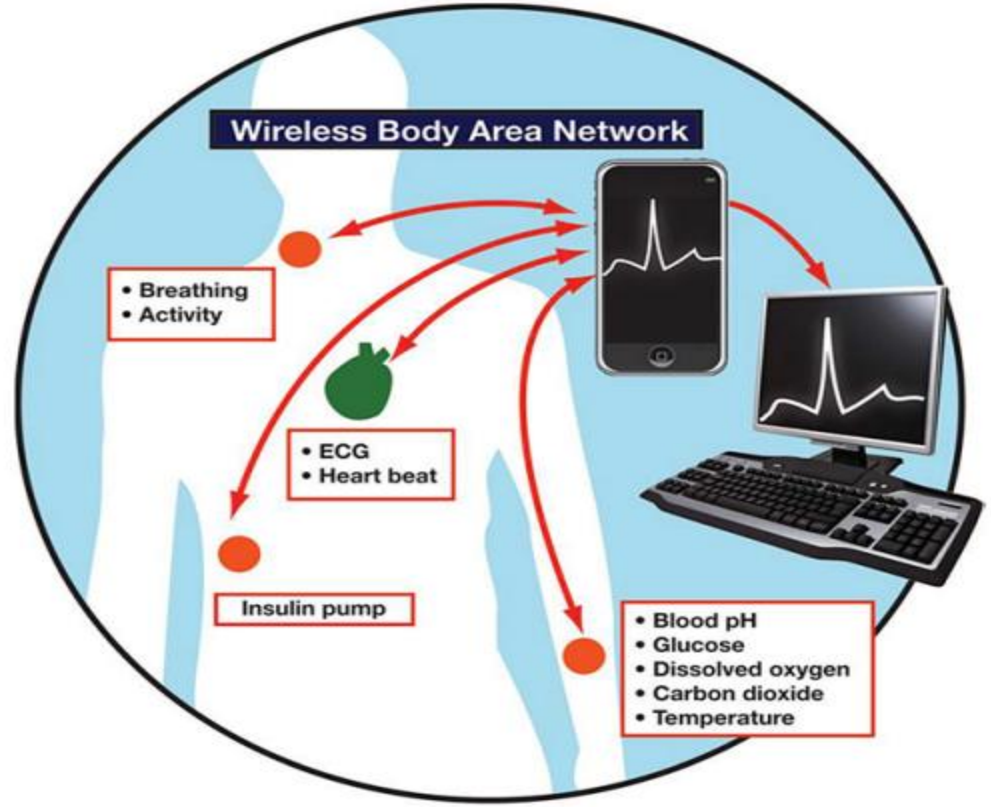


Figure 12.6. *A BAN (Body Area Network)*

Machine learning in IoT

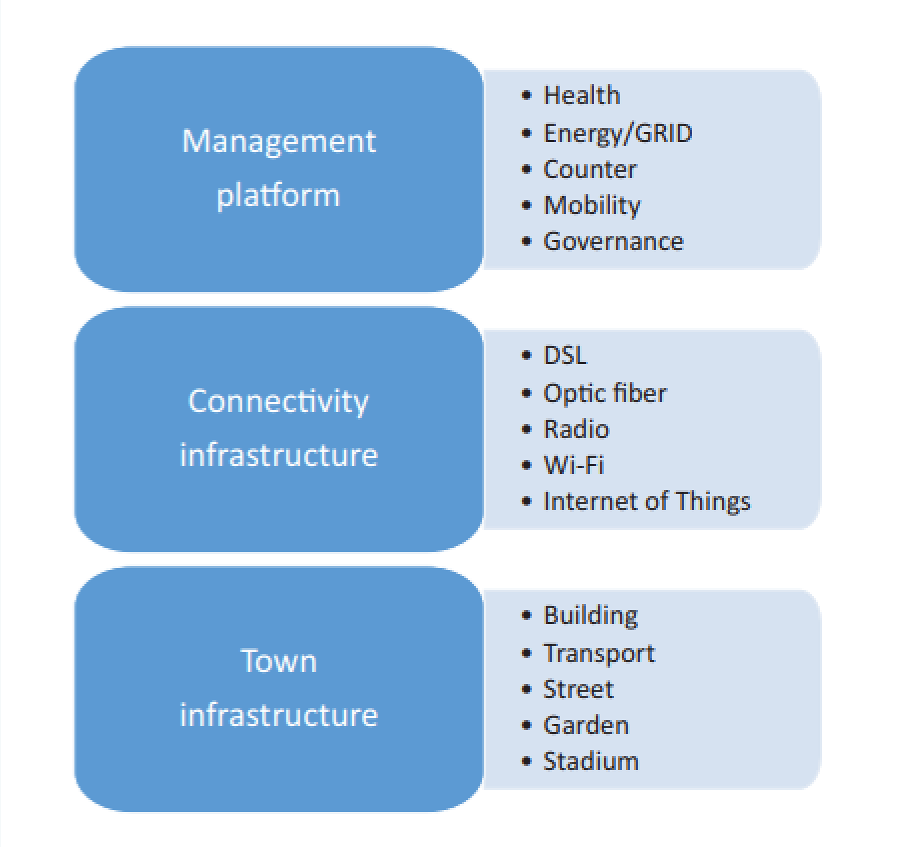
- **NEST Thermostat.** NEST “learns” the regular temperature preferences of its users, and also adapts to the work schedule of its users by turning down energy use. The input is the temperature preference of the user, time and day, presence of the user at home, etc. and the output is a discrete set of temperatures making this a classification problem.
- **Tesla cars.** Tesla enabled auto pilot service in its cars that helps in hands-free driving, including complex tasks like lane changes.
- Tesla cars built since 2014 have 12 sensors on the bottom of the vehicle, a front-facing camera next to the rear-view mirror, and a radar system under the nose.
- These sensing systems are not only constantly collecting data to help the autopilot work on the road, but also to amass data that can make Tesla's operate better in the future.

Smart cities

- Smart cities and transportation: Integration of security services. Optimization of public and private transportation. Parking Sensors.
- Smart management of parking services and traffic in real time. Smart management of traffic lights depending on traffic queues. Locate cars that have overstayed Smart energy grids.
- Security (cameras, smart sensors, information to citizens).
- Water management. Parks and Gardens irrigation. Smart garbage cans. Pollution and mobility controls. Get immediate feedback and opinions from citizens.
- Smart governance. Voting Systems. Accident monitoring, emergency actions coordination.

Sensitivity: Internal

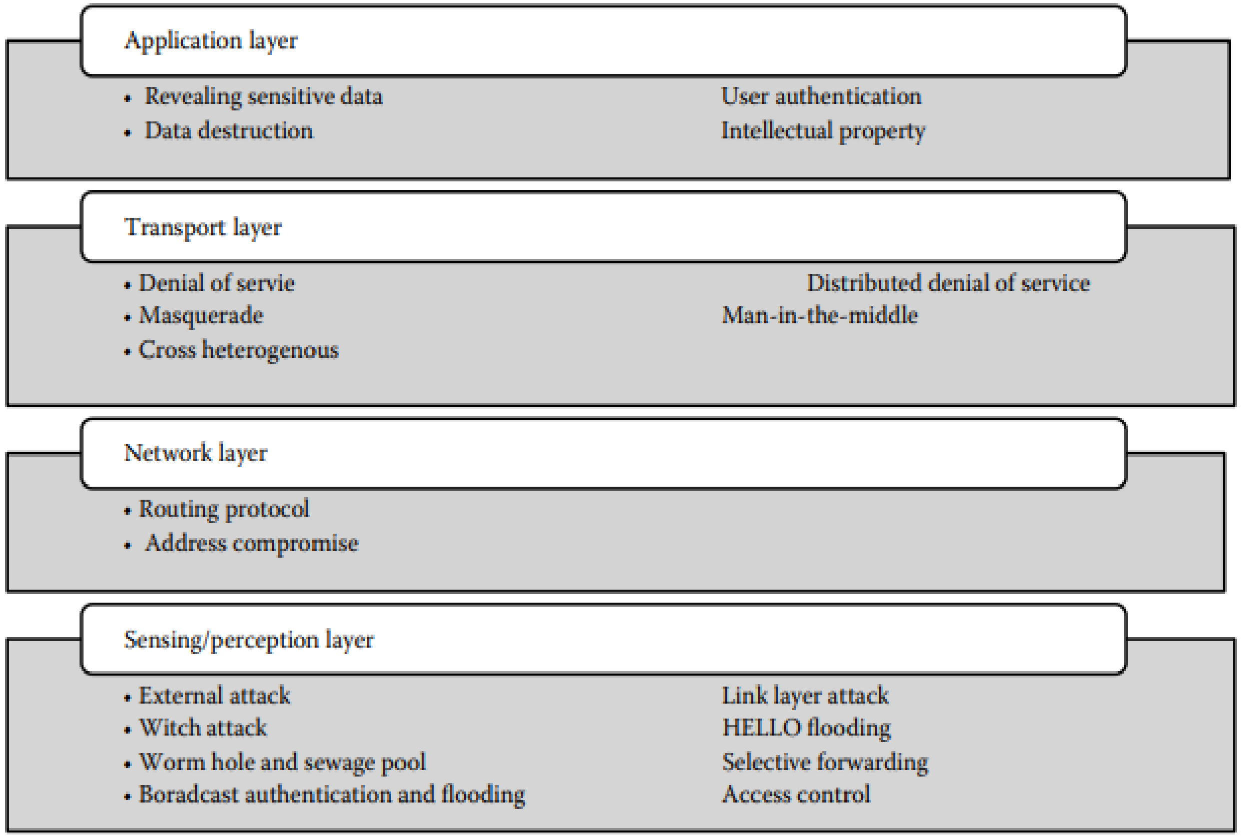
Infrastructure of a Smart city



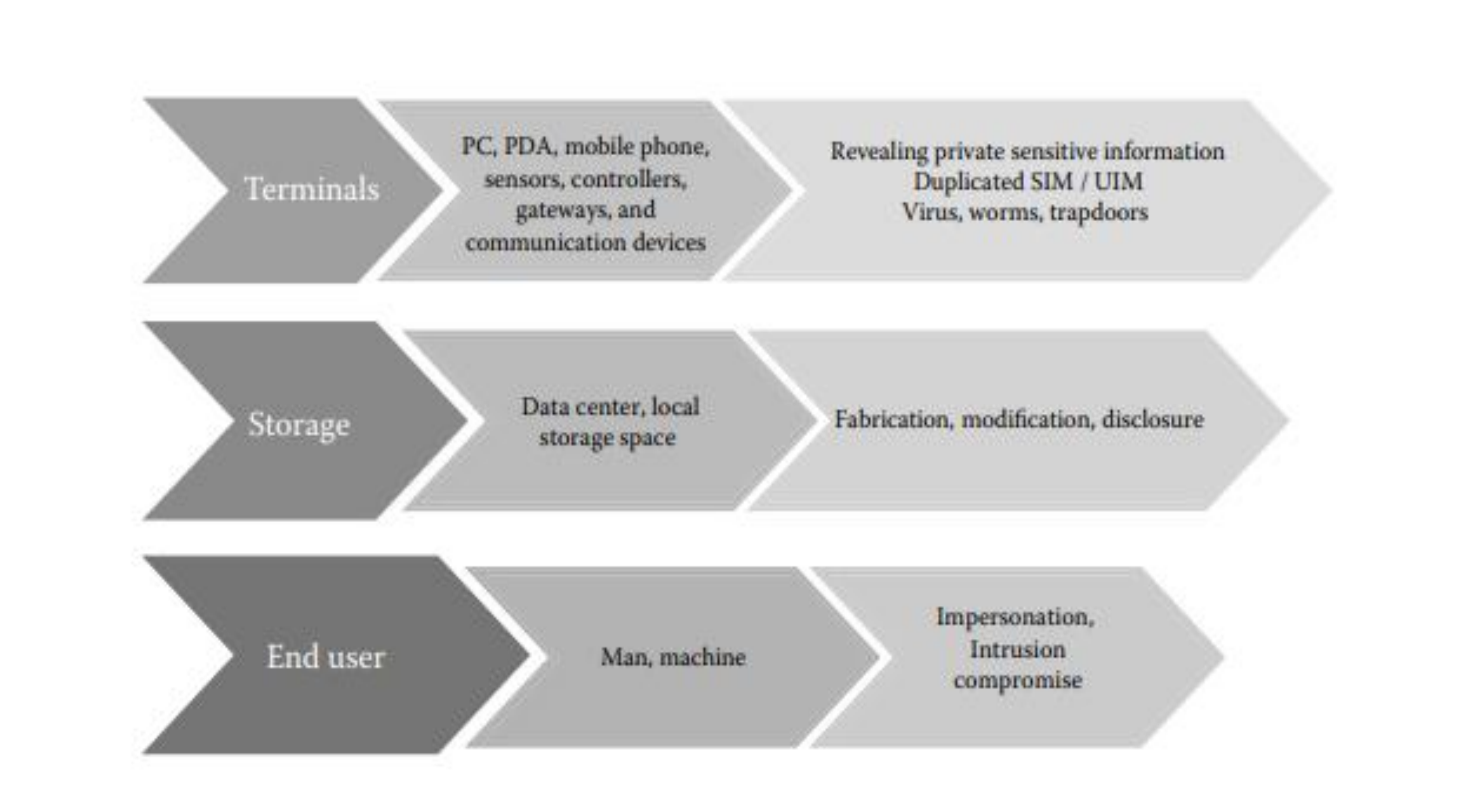
Section 4

IoT Security and Solutions

Attacks based on Layered Architecture



Attacks based on Components



Solutions to IoT Security

- Here are the top four actions recommended by US-CERT to protect IoT devices from attacks:
  - Ensure all default passwords are changed to strong passwords. (Default usernames and passwords for most devices can easily be found on the Internet, making devices with default passwords extremely vulnerable.) for example, it is common knowledge that the default username/password combination for a raspberry pi is **raspberry/pi**
  - Update IoT devices with security patches as soon as patches become available.
  - Disable Universal Plug and Play (UPnP) on routers unless necessary.
  - Purchase IoT devices from companies with a reputation for providing secure devices.

- The application of machine learning (ML) is very critical to IoT security due to the volume and variety of data. AI and ML are fast-growing fields and IoT data analysis needs to be on par with the latest trends in these areas.

Summary

- Introduction to IoT: The Internet of Things (IoT) refers to a network of interconnected physical devices that communicate using sensors and digital connectivity.
- IoT enables machine-to-machine (M2M) communication and data sharing, significantly enhancing automation and smart solutions.
- IoT Architecture and Communication Technologies: IoT follows a four-layer architecture: Object sensing, Data exchange, Information integration, and Application service layers.
- Communication technologies include short-range (Bluetooth, RFID, NFC, ZigBee, Wi-Fi), medium-range (LTE-Advanced), and long-range networks (VSAT, LPWAN), as well as wired options like Ethernet.
- IoT Protocols, Sensors, and Actuators: IoT uses multiple protocols across layers: MQTT (lightweight transport), CoAP (HTTP integration), DTLS (security), and NanoIP (lightweight networking for embedded systems).Sensors (RFID, NFC, LiDAR, Smart Meters) and actuators (hydraulic, pneumatic, electrical) are essential for IoT applications.
- IoT Applications and Use Cases: Industrial IoT (IIoT) is widely applied in smart energy systems (Landis Smart Meter, Nest Thermostat), self-driving vehicles (LiDAR), healthcare (Body Area Networks), and smart cities (traffic management, security, waste management).IoT helps optimize infrastructure, governance, and everyday services by using real-time monitoring, automation, and AI-driven analytics.
- Security Challenges in IoT: IoT is vulnerable to cyber threats such as data breaches, identity theft, network intrusions, and attacks targeting architecture layers (e.g., sensor spoofing, DDoS, malware).Security concerns arise due to limited computational resources in IoT devices, which restrict encryption and authentication mechanisms.

Sensitivity: Internal